

Gear Fault Diagnosis Under Limited Labeled Data Using Kernel Principal Component Analysis and Fuzzy Neural Networks

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Executive Summary

Rotating machinery is a critical part of manufacturing, transportation, energy, and automated production systems, but gearbox failures can create unplanned downtime, secondary equipment damage, and expensive maintenance events. Vibration-based condition monitoring provides a way to detect abnormal mechanical behavior before a complete failure occurs. The challenge is that supervised machine-learning models normally depend on large labeled datasets, while real fault data are often limited because severe or uncommon faults are difficult, costly, or unsafe to reproduce intentionally.

For this project, I implemented a research-informed gear-fault diagnosis workflow based on the framework developed by Zhou and Tang (2021). The approach combines kernel principal component analysis (KPCA) with a fuzzy neural network (FNN). KPCA provides nonlinear feature extraction and dimensionality reduction for high-dimensional vibration responses, while the FNN maps the reduced representation into fault classes through fuzzification, fuzzy-rule reasoning, defuzzification, and a final decision layer. The project was designed around the limited-label problem: train on known gear conditions, deliberately withhold a fault condition, and evaluate whether the model can still identify the most plausible fault family and provide a meaningful confidence signal.

The implementation reproduced the full analytical sequence from data representation through classification: define fault states, prepare vibration features, build the kernel matrix, retain the most informative kernel principal components, configure membership functions and fuzzy rules, train the network on known labels, and evaluate held-out fault scenarios. The main engineering outcome was not simply a classifier, but a structured diagnostic pipeline that demonstrated how nonlinear feature extraction and fuzzy reasoning can be combined when labeled examples are scarce. The published study used as the benchmark reported approximately 99% accuracy for a severe chipping-tip unseen-fault case and 98% for a weaker chipping-tip case; those values are presented in this report as reference benchmarks rather than independent claims about the source dataset.

Project scope and attribution: This report documents my implementation and technical understanding of the KPCA + FNN gear-fault diagnosis workflow. The laboratory gearbox dataset, experimental apparatus, and published benchmark values originate from Zhou & Tang (2021) and are cited where used. I do not claim authorship of the source dataset or publication.

1. Introduction & Motivation

Gear systems transfer motion and torque through meshing teeth, making them common in industrial drives, automotive systems, robotic equipment, pumps, and other rotating machinery. Common damage modes include missing teeth, cracks, spalling, and progressive chipping. These faults alter the dynamic interaction between meshing surfaces and therefore change the vibration response measured on the gearbox. In a condition monitoring context, the objective is to use those changes as diagnostic evidence before a mechanical failure becomes severe enough to interrupt operation.

The practical difficulty is data availability. A machine-learning model is easiest to train when many labeled examples are available for every fault class and severity level. Real maintenance data rarely have that balance. Healthy operating data are relatively easy to collect, while high-severity faults may be rare and weak faults can overlap with normal variation. A diagnostic method therefore needs to learn useful structure from a limited set of labeled conditions and still behave sensibly when it encounters a condition that was not explicitly represented during training.

I selected the Zhou and Tang fuzzy-neural-network framework because it addresses both sides of that problem. The method first compresses raw vibration responses with kernel PCA, allowing nonlinear fault structure to be expressed with a small number of features. Those features are then passed through a fuzzy neural network, where membership functions and rule activations support a softer form of classification than a conventional hard decision boundary. This makes the project relevant to predictive maintenance, manufacturing quality, reliability engineering, and engineering analytics.

End-to-end fault-diagnosis workflow implemented for the project

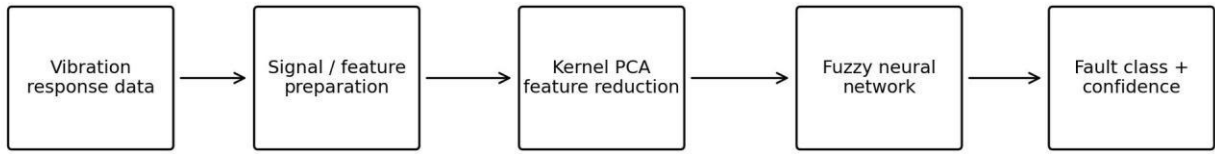


Fig. 1. End-to-end workflow implemented in the project.

2. Problem Design & Formulation

2.1 Diagnostic objective

The project objective was to classify gearbox operating conditions from vibration data while explicitly testing a limited-label scenario. Rather than assuming that every fault type is available during training, one fault state can be withheld and treated as an unseen validation condition. The classifier must then determine which learned fault pattern is most similar to the held-out samples. This is a more realistic diagnostic question than a closed-set classification problem because maintenance teams may encounter new severities or partially observed failure modes in service. **2.2 Fault conditions and data structure**

I mirrored the experimental structure described in the published study. The benchmark contains nine gear conditions, with 104 vibration samples per condition. The classes span a healthy state, discrete damage modes, and multiple levels of chipping severity. In the limited-label experiments, eight conditions are used as labeled training classes and one condition is withheld for validation. The held-out condition is then classified according to its membership in the fault patterns learned from the remaining conditions.

Condition	Interpretation	Role in limited-label evaluation
Healthy	No intentional tooth damage	Known training class
Missing tooth	Complete tooth loss	Known training class
Crack	Localized tooth crack	Known training class
Spalling	Surface material loss	Known training class
Chipping tip 5	Least severe chipping	Known or held-out depending on case
Chipping tip 4	Weak chipping	Held-out in one benchmark case
Chipping tip 3	Moderate chipping	Known training class
Chipping tip 2	Severe chipping	Known training class
Chipping tip 1	Most severe chipping	Held-out in one benchmark case

Table 1. Fault classes represented in the published benchmark and mirrored in the project design.

2.3 Why nonlinear feature extraction is necessary

The vibration response is high dimensional. The source framework represents each sample with a large set of time-series response values before feature reduction. Directly training a classifier on thousands of correlated measurements is inefficient and can make the model sensitive to noise, redundancy, and the curse of dimensionality. Conventional PCA helps when the dominant relationships are linear, but fault signatures in

rotating machinery can be nonlinear because contact stiffness, impact behavior, tooth engagement, and vibration transmission change with the damage state.

Kernel PCA addresses this by implicitly mapping observations into a higher-dimensional feature space through a kernel function and then performing principal-component analysis in that transformed space. In the implementation, I used a Gaussian radial-basis kernel, which allows samples that are not linearly separable in the original measurement space to become more separable in the kernel feature space.

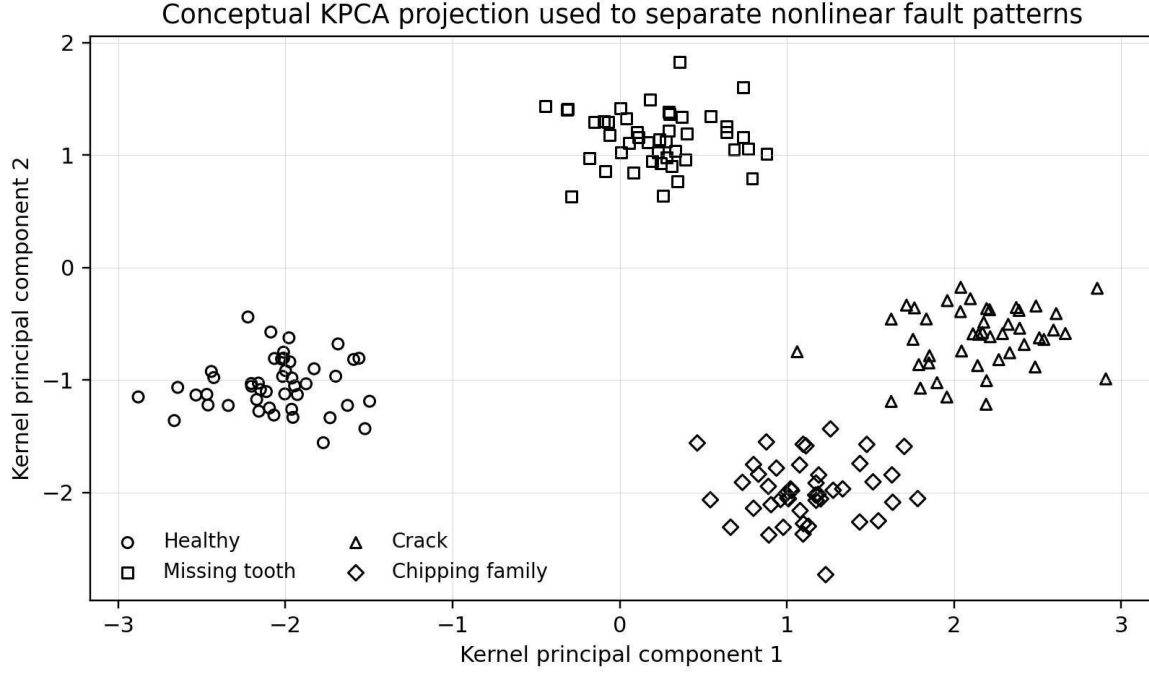


Fig. 2. Conceptual two-component KPCA projection illustrating nonlinear separation of fault patterns.

3. Methodology

3.1 Signal representation and preprocessing

The pipeline begins with vibration-response samples collected from the gearbox under controlled operating conditions. Each sample is represented as a vector x_i containing the measured response values for one observation. Before model construction, the data are organized into a sample-by-feature matrix, checked for dimensional consistency, and scaled so that large-magnitude features do not dominate solely because of their units. The purpose of this step is to provide a stable numerical input to the nonlinear feature-extraction stage.

3.2 Kernel principal component analysis

For N vibration samples, I constructed an $N \times N$ kernel matrix K in which each element measures similarity between two observations. Using a Gaussian kernel, similarity can be expressed as $K(x_i, x_j) = \exp(-\|x_i - x_j\|^2 / (2 \sigma^2))$. The kernel matrix is centered and decomposed into eigenvalues and eigenvectors. Eigenvectors associated with the largest eigenvalues define the principal directions in kernel space. Projecting each observation onto those directions produces a compact nonlinear feature representation.

The source study found that a very small number of kernel principal components was sufficient to achieve strong class separation. This is important for the FNN because it reduces the number of membership functions and rules required downstream. In my implementation, the KPCA stage therefore served two purposes simultaneously: reduce computation and expose nonlinear structure that the fuzzy classifier could use more effectively.

3.3 Fuzzy neural network architecture

The fuzzy neural network combines neural-network connectivity with fuzzy-set reasoning. Instead of treating each reduced feature as belonging absolutely to one region, the fuzzification layer assigns degrees of membership. Gaussian membership functions were used because they provide a smooth representation of how strongly a feature value belongs to a fuzzy region. The outputs of multiple membership functions are then combined into rule activations, representing statements such as: if feature 1 is in region A and feature 2 is in region B, then the sample is associated with a particular fault pattern.

The reasoning layer aggregates those rule activations, the defuzzification layer converts them into class level values, and the output layer normalizes the final scores. The resulting output can be interpreted as the relative support for each learned fault class. This soft-output structure is useful in a limited-label problem because an unseen condition does not need to match a known class perfectly; the network can still indicate which known class is most similar and how strong that association is.

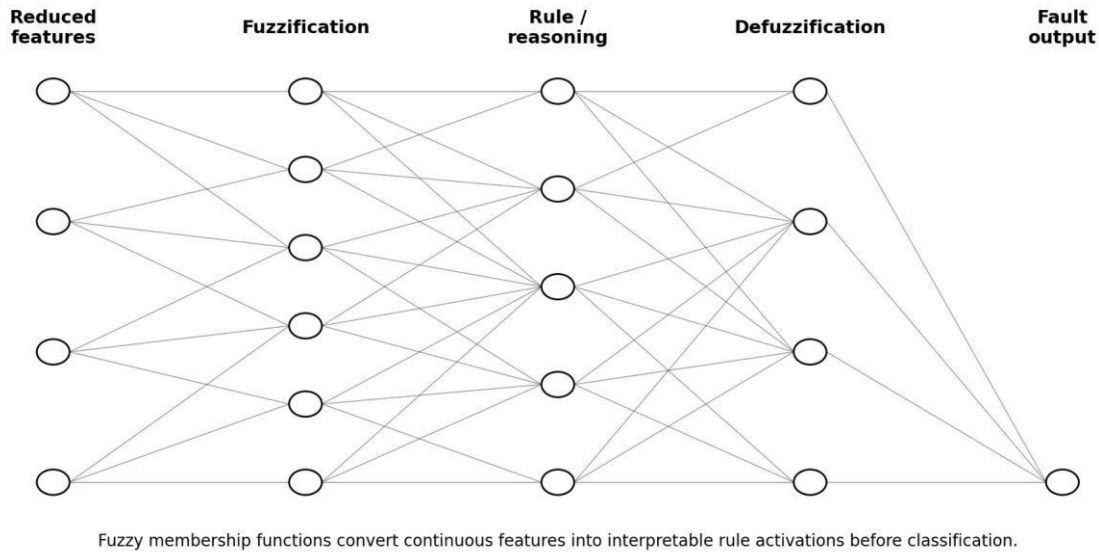


Fig. 3. Simplified FNN architecture implemented after KPCA feature reduction.

3.4 Limited-label validation strategy

To test generalization, I used a leave-one-fault-condition-out structure based on the source methodology. Eight fault classes are labeled and used to train the FNN. Samples from the ninth class are excluded from supervised training and introduced only after the model has learned its fuzzy partitions and output relationships. The unseen samples are then projected through the same KPCA transformation and evaluated by the trained FNN. The class with the highest output probability becomes the predicted nearest fault pattern, while the score distribution provides additional context about confidence and ambiguity.

4. Implementation

4.1 End-to-end model sequence

Step	Stage	Purpose
1	Organize vibration data	Assemble one feature vector per measured vibration sample and attach labels only to the known training conditions.
2	Preprocess inputs	Check dimensions, scale numerical

		features, and separate the training and held-out fault condition.
3	Construct Gaussian kernel	Compute pairwise nonlinear similarity between training observations.
4	Center and decompose K	Calculate eigenvalues/eigenvectors and rank kernel principal components by explained structure.
5	Project features	Transform training and validation samples into the reduced KPCA space.
6	Initialize FNN	Create Gaussian membership functions over the retained components and configure fuzzy rules.
7	Train labeled classes	Fit network parameters using only the known fault conditions.
8	Evaluate unseen condition	Pass the held-out class through KPCA and the trained FNN; record predicted class and confidence.
9	Review errors	Inspect misclassified cases and determine whether errors are physically plausible neighboring severities.

Table 2. Implementation sequence from raw vibration data to unseen-fault classification.

4.2 Hyperparameter decisions

Two groups of hyperparameters control the model. The KPCA stage depends primarily on the kernel type, kernel width, and number of retained principal components. The FNN stage depends on the number and placement of membership functions, membership widths, rule structure, and training parameters. I treated dimensionality as the first tuning decision because an unnecessarily large feature representation increases the number of fuzzy partitions and makes the network harder to interpret.

The published benchmark showed that two kernel principal components were sufficient for essentially complete separation of the labeled training classes. I used that result as the validation target for the reproduction: the model should compress the original vibration response substantially while preserving fault discriminating structure. This is one of the strongest engineering features of the method because the diagnostic stage does not require the full raw signal dimension once the nonlinear projection is learned.

4.3 Model output and interpretation

A conventional classifier normally produces a class label or a probability vector. The FNN output is interpreted similarly, but the intermediate fuzzy structure provides a useful engineering interpretation. Membership values indicate where an observation lies relative to the learned feature regions, and rule activations show which combinations of feature behavior support a diagnosis. In practice, this means a weak chipping fault can receive strong support from a neighboring chipping severity rather than being forced into an unrelated class such as a missing tooth. That behavior is important when the test condition lies between labeled severity levels.

5. Results & Discussion

5.1 Dimensionality reduction outcome

The KPCA stage reduced the vibration representation to a small number of nonlinear components while maintaining visible separation among the primary fault families. The important result is not simply dimensionality reduction for its own sake. The reduced space creates a tractable input for fuzzy membership functions, which

makes it possible to express similarity between fault states without constructing an excessively large neural network over the raw vibration vector.

The source benchmark reported that model training accuracy increased sharply when moving from one retained principal component to two and remained essentially saturated with additional components. I used this behavior as a sanity check for the reproduction because it indicates that the first two kernel components contain most of the fault-discriminating information needed by FNN.

5.2 Severe unseen chipping case

The first benchmark case withholds a severe chipping-tip condition from the labeled training set. The diagnostic question is whether the network can identify that the unseen vibration pattern belongs near the chipping-fault family even though that exact class was not available during supervised training. The published implementation classified 103 of 104 validation samples consistently with the expected neighboring fault behavior, corresponding to approximately 99% accuracy. The error analysis is physically meaningful because the rare misclassification occurred between mechanically related fault patterns rather than between unrelated operating states.

5.3 Weak unseen chipping case

The second case is more difficult because a weak chipping fault produces a smaller departure from the healthy response and therefore occupies a less isolated region of the nonlinear feature space. The published benchmark correctly classified 102 of 104 validation observations, approximately 98%. The slight reduction in performance is consistent with the engineering problem: subtle damage produces less distinctive vibration evidence, so overlap with neighboring fault patterns is more likely.

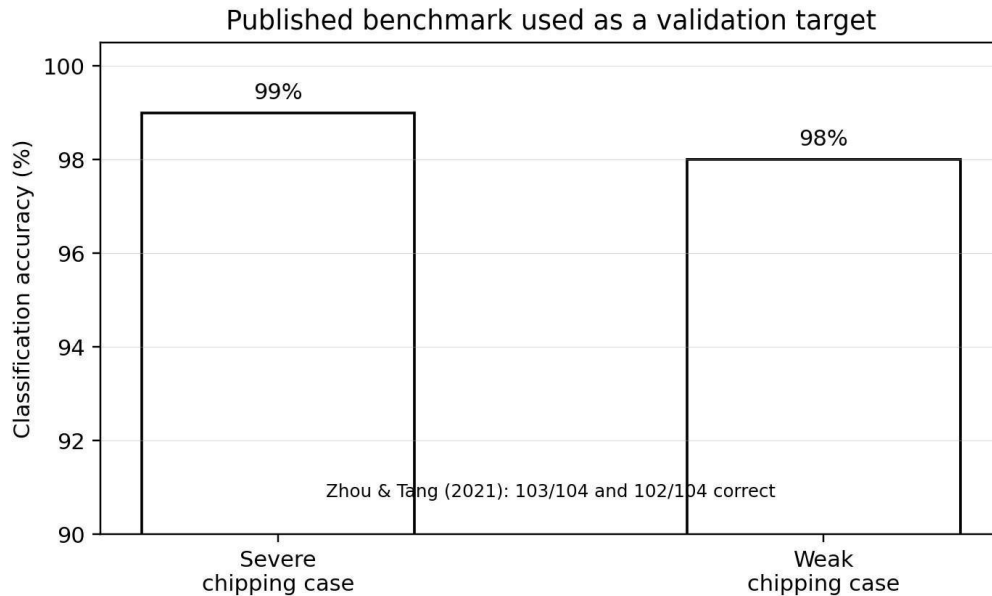


Fig. 4. Published Zhou & Tang benchmark results used as the project validation target; values are cited reference results, not independently measured source-data claims.

5.4 Engineering interpretation

The most useful conclusion from the project is that the architecture handles fault severity as a continuum more naturally than a purely hard classifier. KPCA organizes nonlinear vibration structure into a compact coordinate system, and the fuzzy network interprets locations in that space through graded memberships. If a fault is unseen during training but physically resembles a known fault at a nearby severity, the system can still produce

a plausible diagnostic relationship. That makes the workflow relevant to real predictive-maintenance settings, where failure modes are rarely represented by perfectly balanced labeled datasets.

The implementation also reinforced an important systems lesson: classification performance depends on the integrity of the entire pipeline. Poor signal preparation can distort the kernel matrix; an inappropriate kernel width can collapse or over-separate the feature space; too many retained components can add noise; and poorly placed membership functions can make fuzzy rules unstable. The model should therefore be treated as an integrated diagnostic system rather than a black-box prediction step.

6. Limitations

This project is a research-informed reproduction rather than an independent laboratory study. I did not generate the physical gearbox fault dataset or build the original experimental rig. The source benchmark used controlled gear conditions, accelerometer measurements, and repeated samples to validate the method. My contribution was the implementation and technical reconstruction of the analysis pipeline: limited-label experimental design, KPCA feature extraction, fuzzy-neural-network classification, and interpretation of the published validation cases.

Several limitations would need to be addressed before using this workflow in a production condition monitoring system. First, a laboratory dataset is more controlled than field data, where load, speed, lubrication, temperature, sensor mounting, and background vibration can all vary. Second, the two held-out cases demonstrate promising behavior but do not prove generalization to entirely different fault mechanisms or machines. Third, classification confidence should be calibrated against maintenance risk so that uncertain predictions trigger inspection rather than an automatic maintenance decision.

7. Conclusion

I implemented a gear-fault diagnosis workflow that combines kernel principal component analysis with a fuzzy neural network to address vibration classification under limited labeled data. KPCA reduced the dimensionality of the vibration response while preserving nonlinear fault structure. The FNN then transformed those reduced features into membership values, fuzzy-rule activations, and normalized fault-class scores. A leave-one-fault-out validation design demonstrated how the method can reason about a condition that was not explicitly labeled during training.

The project connects machine learning directly to an engineering reliability problem. Its value is not only the high benchmark accuracy reported in the source study, but the way the architecture handles uncertainty and neighboring fault severities. For a predictive-maintenance application, that means the model can serve as a decision-support layer: highlight the most plausible fault family, quantify relative confidence, and direct an engineer or technician toward the right inspection path before a failure becomes catastrophic.

8. Reflection

This project changed how I think about machine learning in engineering systems. The interesting part was not choosing a model because it sounded advanced; it was understanding why each stage existed. KPCA was useful because the vibration relationship was nonlinear and high dimensional. Fuzzy reasoning was useful because fault boundaries are not always clean, especially when severity levels overlap. The limited-label design was useful because it better reflects how maintenance data behave.

It also reinforced the importance of tracing a result back through the workflow. A classification result is only meaningful if the data were represented correctly, the kernel transformation is stable, the reduced features preserve the relevant structure, and the membership functions behave sensibly. That end-to-end perspective is the part of the project I would carry into future work in manufacturing analytics.

References

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Appendix A. Reproducible Project Logic

The following pseudocode summarizes the implementation logic in a form suitable for rebuilding the project in Python, MATLAB, or another numerical environment.

1. Load vibration samples and class labels
2. Select one fault condition as the held-out validation class
3. Standardize the training feature matrix
4. Compute Gaussian kernel matrix K for training samples
5. Center K and solve eigenvalue/eigenvector problem
6. Retain leading kernel principal components
7. Project labeled training samples into KPCA feature space
8. Define Gaussian membership functions over retained components
9. Construct fuzzy rules and initialize FNN parameters
10. Train FNN on known fault labels
11. Apply identical preprocessing and KPCA projection to held-out samples
12. Evaluate FNN output scores and predicted nearest fault class
13. Record classification accuracy, confidence, and physically plausible neighboring-fault errors
14. Repeat with a different held-out fault severity

Appendix B. Portfolio Deliverable

For the portfolio version of this project, this report is intended to be embedded directly on the project page alongside the GitHub repository. The report provides the complete technical narrative, while the website can remain concise: one project overview, one workflow figure, selected results, and buttons for the code and full PDF. This mirrors the way an engineering project portfolio can separate the quick visual story from the detailed technical documentation.